

Introduction to Data Science (DS2001)

Date: Nov 6th 2025

Course Instructor(s)

Ms. Mahrukh Khan

Sessional -2

Total Time (Hrs): 1

Total Marks: 15

Total Questions: 3

Semester: Fall

2025

Dept: AI&DS

Roll No

Section

Student Signature

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Attempt all questions on the answer script.

Show all computations.

CLO #3: Students should be able to apply programming languages and open-source tools to manipulate data, develop predictive models, and visualize results effectively.

Question#1:

A KNN model with $K=1$ gives Training accuracy = 100% and Test accuracy = 72%.

A KNN model with $K=25$ gives Training accuracy = 85% and Test accuracy = 83%.

- (a) Which model overfits and why? [2 marks]
- (b) Explain how K influences bias and variance. [2 marks]
- (c) What value of K would you prefer and why? [1 mark]

a) **$K = 1$ gives perfect training accuracy but poor test accuracy → the model memorizes training data and fails to generalize. → This is overfitting.**

$K = 25$ performs consistently on both training and test data → better generalization.

b) **Small K (e.g., $K=1$):**

Low bias (fits data very closely)

High variance (small change in data → big change in prediction)

Overfits the training data

Large K (e.g., $K=25$):

High bias (decision boundary smoother)

Low variance (less sensitive to noise)

May underfit if K is too large

c) Prefer **$K = 25$** , because it gives a **balanced bias–variance trade-off** and performs better on unseen data.

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CLO #2: Students should be able to design and implement data pre-processing pipelines to clean, transform, and prepare raw data for analysis, ensuring data quality and integrity.

Question#2(A):

Define the following metrics and state when each is more useful, include atleast one application for each. [2 marks]

1. Precision
2. Recall
3. F1-score
4. Accuracy

- **Precision:**
Of all items predicted *positive*, how many are actually *positive*? Useful when **false positives** are costly (e.g., spam detection — don't want to mark real mail as spam).
- **Recall:**
Of all actual *positives*, how many were correctly predicted? Useful when **false negatives** are costly (e.g., disease detection — don't miss a positive case).
- **F-1 Score:**
Harmonic mean of precision and recall. Useful when you want a **balance** between precision and recall.
- **Accuracy:**
Proportion of total correct predictions. Useful when **class distribution is balanced** (not skewed).

Question#2(B):

Compute the above-mentioned metrics for the confusion matrix given below. [4 marks]

	Predicted Positive	Predicted Negative
Actual Positive	40	10
Actual Negative	5	45

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$$\text{Precision} = \frac{TP}{TP + FP} = \frac{40}{40 + 5} = \frac{40}{45} = 0.8889$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{40}{40 + 10} = \frac{40}{50} = 0.8$$

$$F1 = 2 \times \frac{(0.8889)(0.8)}{0.8889 + 0.8} = 2 \times \frac{0.7111}{1.6889} = 0.842$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{40 + 45}{40 + 45 + 5 + 10} = \frac{85}{100} = 0.85$$

CLO #1: Students should be able to articulate a comprehensive understanding of Data Science, encompassing its fundamental concepts, methodologies, and practical applications.

Question#3: A study shows:

- 45% of students prefer shopping online,
- 55% prefer shopping in-store.
Among online shoppers, 70% say they overspend.
Among in-store shoppers, 20% say they overspend.

If a student says they overspend, what is the probability they are an online shopper? [4 marks]

Given:

$P(O)=0.45$ = Probability student prefers **online** shopping

$P(S)=0.55$ = Probability student prefers **in-store** shopping

$P(\text{overspend} \mid O)=0.70$

$P(\text{overspend} \mid S)=0.20$

We are asked to find $P(O \mid \text{overspend})$

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$$P(O|\text{overspend}) = \frac{P(\text{overspend}|O) \cdot P(O)}{P(\text{overspend})}$$

$$\begin{aligned} P(\text{overspend}) &= P(\text{overspend} | O)P(O) + P(\text{overspend} | S)P(S) \\ &= (0.70)(0.45) + (0.20)(0.55) \\ &= 0.315 + 0.11 = 0.425 \end{aligned}$$

$$P(O|\text{overspend}) = \frac{(0.70)(0.45)}{0.425}$$

$$= \frac{0.315}{0.425} = 0.741$$

$$P(O|\text{overspend}) = 0.741 \text{ or } 74.1\%$$

If a student says they overspend, there is about a 74% chance that they are an online shopper.